Sophie Zhu

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Education

Columbia University School of Engineering and Applied Sciences Expected May 2028
Bachelor of Science, Major in Chemical Engineering, Minor in Artificial Intelligence
- GPA: 3.9/4.0 | Dean's List

Phillips Exeter Academy May 2024
Classical Languages Diploma
- GPA: 10.5/11 | Highest Honors

Work Experience

Research Assistant Jan 2025 – Present
Columbia University Chemical Engineering, Jingguang Chen Group
- Research CO_2-to-carbon-nanofiber conversion as a domestic low-emission alternative to conventional carbon material production, co-authored in *Advanced Energy Materials* (2026) (https://doi.org/10.1002/aenm.202506784)
- Analyze electrochemical performance data, reducing systems costs by 15% and producing publication figures
- Present visualized findings to the group weekly to inform catalyst and reactor testing decisions
- Develop 3+ first-of-kind technoeconomic models linking energy demand to cost competitiveness

Electrolyzer MEA Engineering Intern May 2026 – Aug 2026
Plug Power Inc.
- Developed 15 MW PEM electrolyzer stacks for industrial hydrogen production capabilities
- Assembled electrical and fluid systems for custom test stands with a 5-person team, raising throughput 16%
- Troubleshot daily hardware and instrumentation faults using SOPs and PFDs to protect testing schedules
- Consolidated 500+ Excel data files and Powerpoint reports into searchable database to evaluate testing strategy
- Led ex-situ membrane degradation experiments for electrolyzer durability from design to executive presentation

Energy Engineering Intern Jun 2025 – Nov 2025
Grid Symphony
- Analyzed large government energy datasets in Excel to assess 4 sustainable fuel cell power options
- Authored a 10-page report summarizing market trends and cost performance for leadership review
- Conducted comprehensive technoeconomic models (comparative market analysis, scenario analysis) to inform investment and commercialization decisions for a multi-million-dollar AI data center

Leadership Activities

Mentor Aug 2026 – Present
New York Academy of Sciences
- Coach student team through a 10-week structured problem-solving cycle from scoping to executive presentation

Co-President Oct 2025 – Present
American Institute of Chemical Engineers Columbia University Chapter
- Lead chapter operations, including financial management, conference planning, and faculty coordination
- Organize professional development, tutoring, and networking events for 200+ undergraduate members

Community Values Committee Member Oct 2024 – Present
Columbia University Chemical Engineering Department
- Direct community initiatives including semesterly town halls, laptop lending programs, etc. for 300+ students in a 12 person team of faculty and undergraduate, masters, PhD, and post-doctorate students

Senior Advisor Oct 2024 – Present
Columbia University Chinese Students Club
- Manage team of 10 to arrange and conduct outreach, logistics, and marketing for Lunar Gala event of 200 people

Additional Information

Awards: 2025 Symposium Best Poster (STEM) 2nd Place (200+); 2025 Engineering Alumni Association Fellow (1/200+)
Computer: Python, R, CAD, Revit, Microsoft Office Suite, Photoshop, Indesign, Figma
Interests: Amateur knitting, Balanchine ballet, Chinese-Korean baking, New York City dining, Ancient Greek literature